Provenance-Aware Agentic Knowledge Graph Systems for Reliable Multi-Source Grounded AI

Primary Topic: Knowledge Grounding **Secondary Topic:** Agentic AI

Retrieval-augmented generation (RAG) [1] has become a foundational technique for grounding large language model (LLM) inference in external knowledge. Production systems have matured substantially: hybrid retrieval combining semantic (dense) and keyword (sparse) search is now standard practice [2], and retrieval is increasingly exposed as a tool within agentic workflows where models decide *when* and *what* to retrieve [3].

However, reliability remains a fundamental challenge. Consider a student using an AI research assistant to synthesize literature across multiple papers. While modern RAG systems can retrieve relevant documents, they often fail to provide verifiable reasoning chains explaining how individual claims were derived. This limitation arises from two key gaps. First, even advanced retrieval systems lack structured reasoning substrates capable of multi-hop synthesis across heterogeneous knowledge sources—an issue that graph-based approaches such as GraphRAG [4] begin to address but do not fully resolve. Second, current systems provide limited provenance: they may cite documents but cannot generate fine-grained evidence chains linking specific outputs to the entities, relationships, and source spans that support them, nor can they reliably detect or recover from retrieval failures [5], [6].

This project proposes **Provenance-Aware Agentic Knowledge Graph Systems (PA-AKG)**, a framework that integrates structured knowledge graphs, adaptive retrieval, and agent-based validation to produce verifiable grounded reasoning. PA-AKG constructs a *provenance graph* linking each generated output span to knowledge graph entities, relationships, and source passages with associated confidence scores. Agents dynamically route queries across dense vector search, sparse retrieval, and structured knowledge graph pathways, while validation agents analyze provenance completeness and evidence consistency to detect weak grounding or contradictory signals.

The methodology combines automated claim extraction, provenance-linked knowledge graph construction, and agentic orchestration for retrieval and validation. To evaluate the approach, we will prototype PA-AKG using Amazon Bedrock Knowledge Bases [2] and compare it against vector-based RAG, hybrid retrieval, and GraphRAG-style reasoning across **multi-hop question answering over heterogeneous knowledge sources** and **claim verification with evidence-chain reconstruction**. Evaluation metrics include answer accuracy, provenance completeness, reasoning consistency, and failure recovery under incomplete or conflicting retrieval conditions.

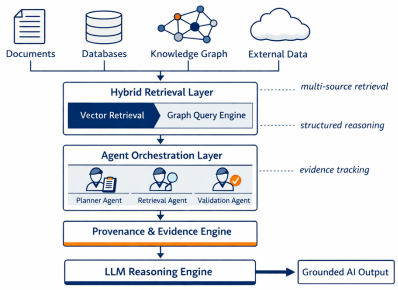

Fig. 1. **PA-AKG Architecture.** Heterogeneous knowledge sources feed a hybrid retrieval layer (vector search + KG queries). An agent orchestration layer coordinates planning, retrieval, and validation agents. A provenance engine records evidence links aligned with observability standards [7] for traceable verification.

Expected outcomes include an open PA-AKG prototype, provenance-aware evaluation benchmarks, and empirical evidence demonstrating that **structured provenance-aware grounding** improves reasoning reliability, traceability, and robustness for agentic AI systems operating over heterogeneous knowledge environments.

References

[1] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W.-t. Yih, T. Rocktäschel, S. Riedel, and D. Kiela, "Retrieval-augmented generation for knowledge-intensive NLP tasks," in *Advances in Neural Information Processing Systems*, vol. 33, 2020, pp. 9459–9474.

[2] Amazon Web Services, "Amazon Bedrock knowledge bases now supports hybrid search," AWS Machine Learning Blog, Mar. 2024, accessed 2026-03-10. [Online]. Available: https://aws.amazon.com/blogs/machine-learning/amazon-bedrock-knowledge-bases-now-supports-hybrid-search/

[3] OpenAI, "Function calling guide," OpenAI Developer Documentation, 2025, accessed 2026-03-10. [Online]. Available: https://platform.openai.com/docs/guides/function-calling

[4] D. Edge, H. Trinh, N. Cheng, J. Bradley, A. Chao, A. Mody, S. Truitt, and J. Larson, "From local to global: A graph RAG approach to query-focused summarization," *arXiv preprint arXiv:2404.16130*, 2024. [Online]. Available: https://arxiv.org/abs/2404.16130

[5] S.-Q. Yan, J.-C. Gu, Y. Zhu, and Z.-H. Ling, "Corrective retrieval augmented generation," *arXiv preprint arXiv:2401.15884*, 2024. [Online]. Available: https://arxiv.org/abs/2401.15884

[6] Z. Ji, N. Lee, R. Frieske, T. Yu, D. Su, Y. Xu, E. Ishii, Y. Bang, A. Madotto, and P. Fung, "Survey of hallucination in natural language generation," *ACM Computing Surveys*, vol. 55, no. 12, pp. 1–38, 2023.

[7] OpenTelemetry, "Semantic conventions for generative AI systems," OpenTelemetry Specification, 2024, accessed 2026-03-10. [Online]. Available: https://opentelemetry.io/docs/specs/semconv/gen-ai/